Ensemble Learning

Training multiple models and combining their predictions.

Boosting

Core Idea

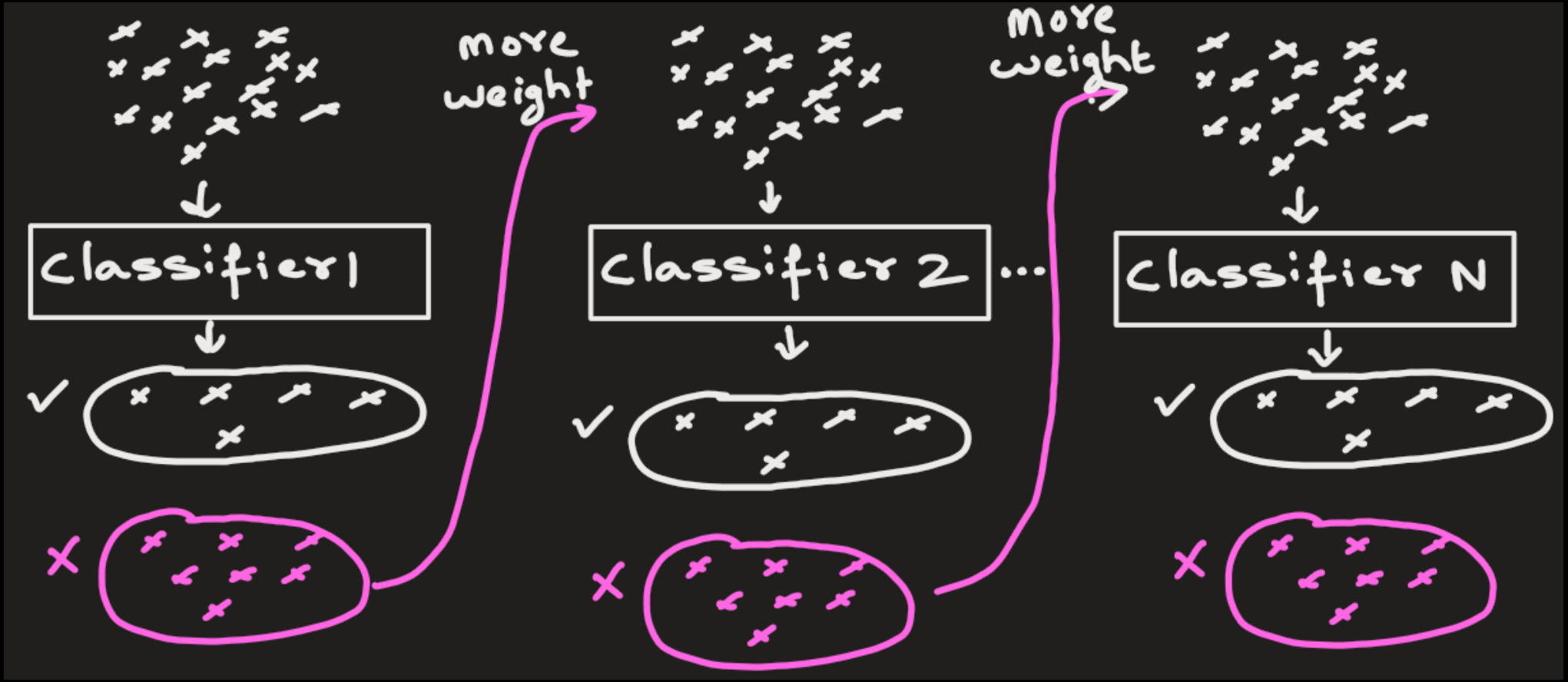
Train models **sequentially**, where each new model focuses on **correcting the mistakes of the previous ones**.

Key Idea: Learn from mistakes step by step

How it Works

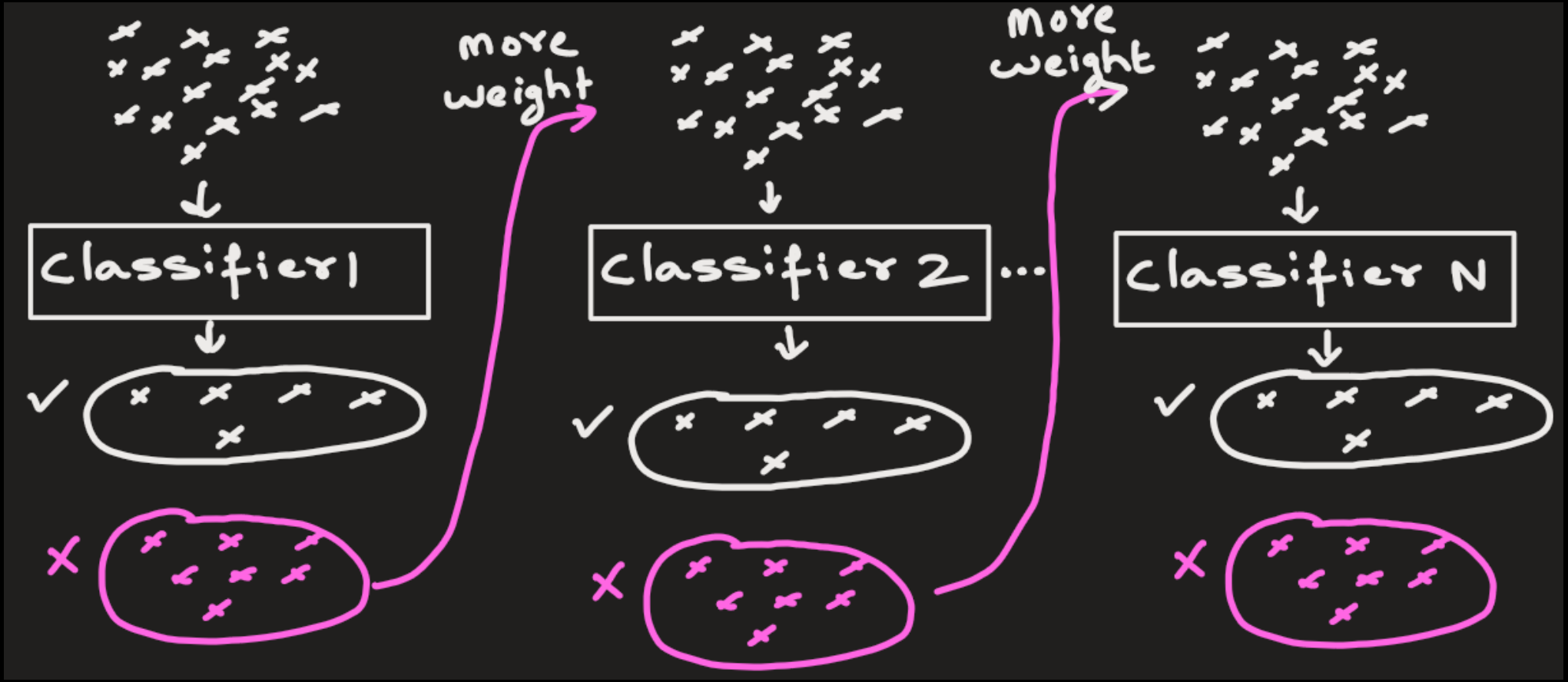
Step 1: Train First Model

- Train a simple model (usually weak learner like decision tree stump).
- Make predictions.
- Identify wrong predictions.



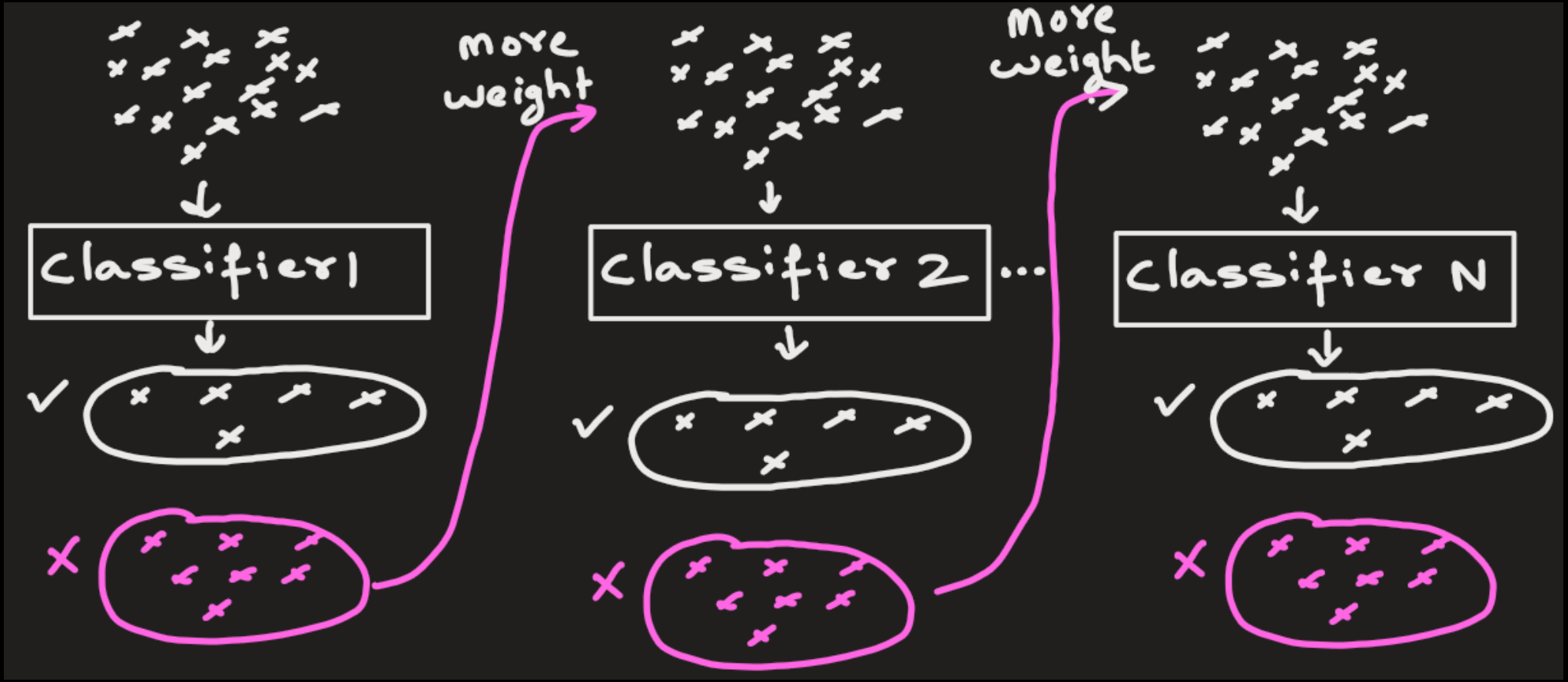
Step 2: Focus on Errors

- Give **more importance (higher weight)** to incorrectly predicted samples.
- Reduce importance of correctly predicted samples.



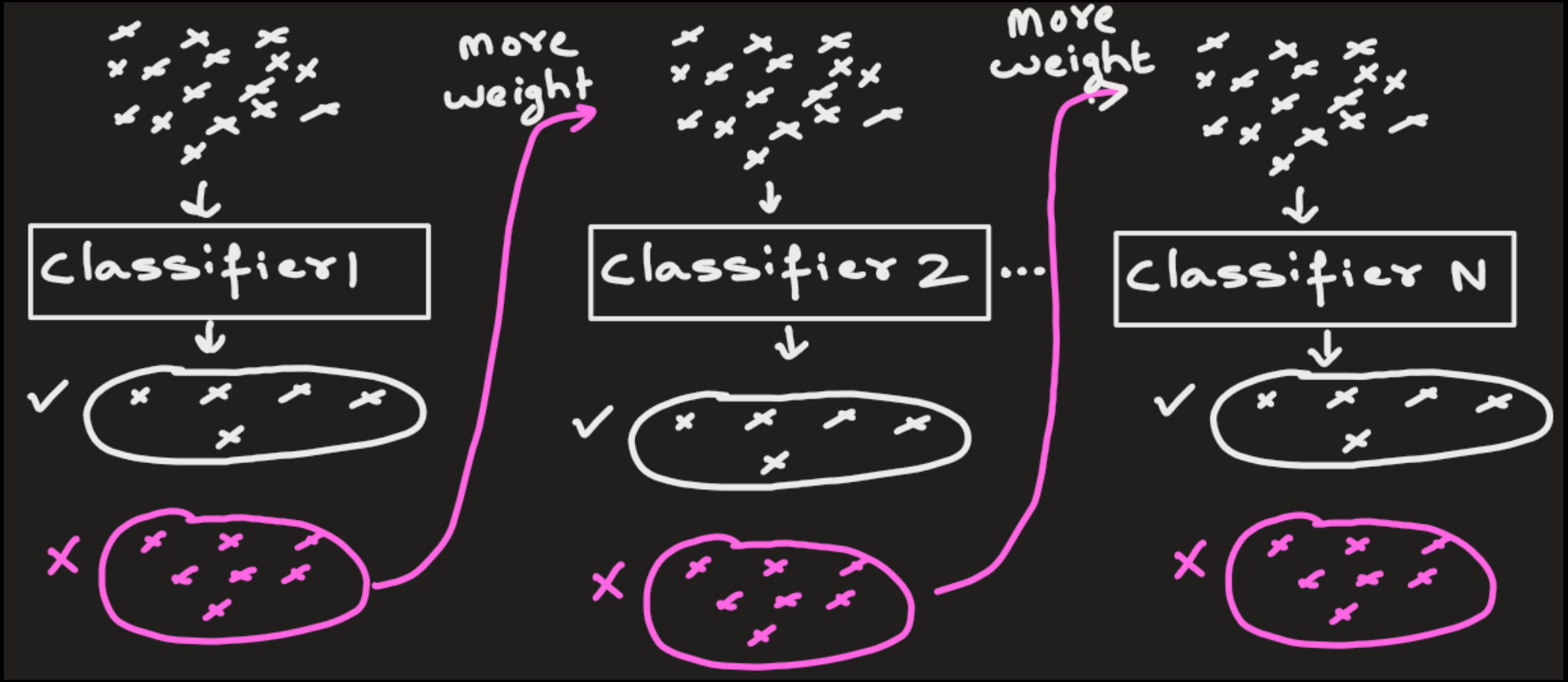
Step 3: Train Next Model

- New model learns mainly from previous mistakes.



Step 4: Combine Models

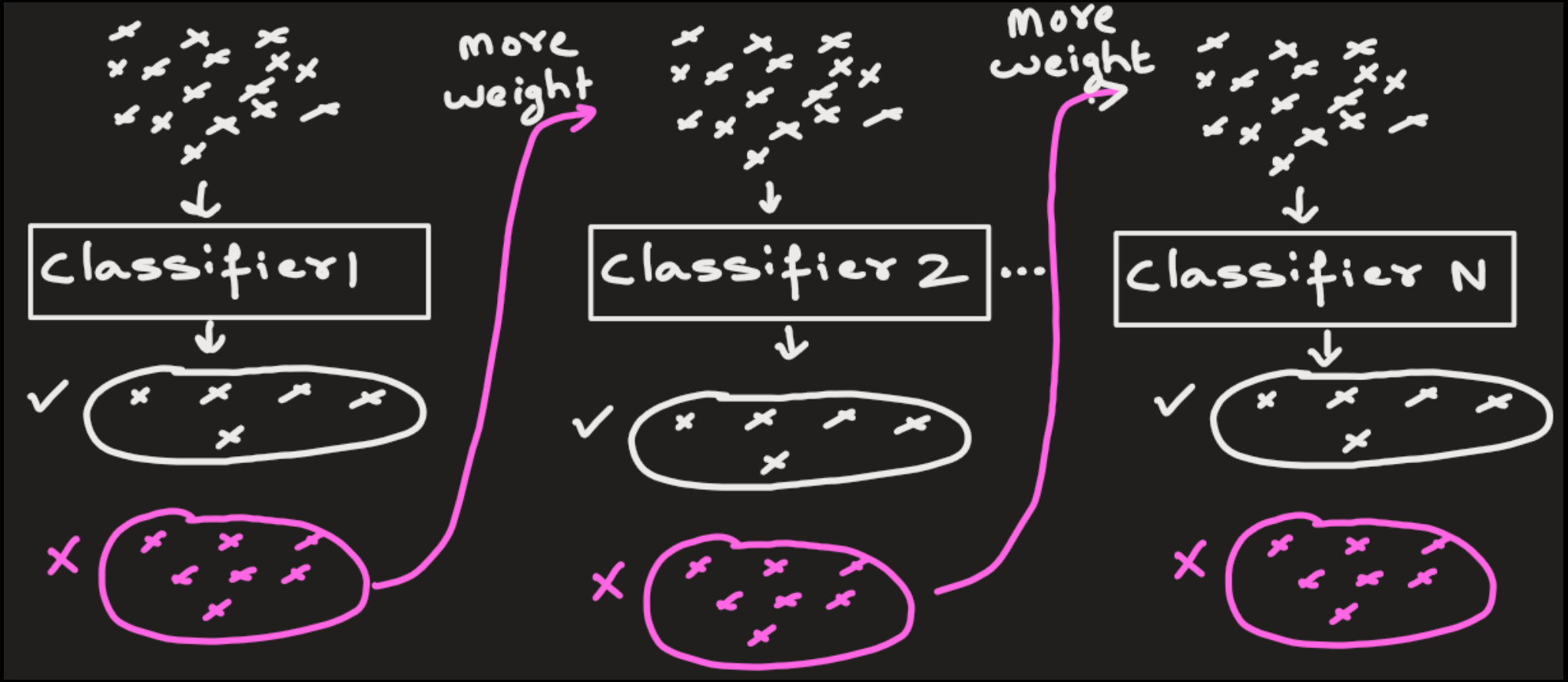
- Final prediction = weighted combination of all models.



Boosting

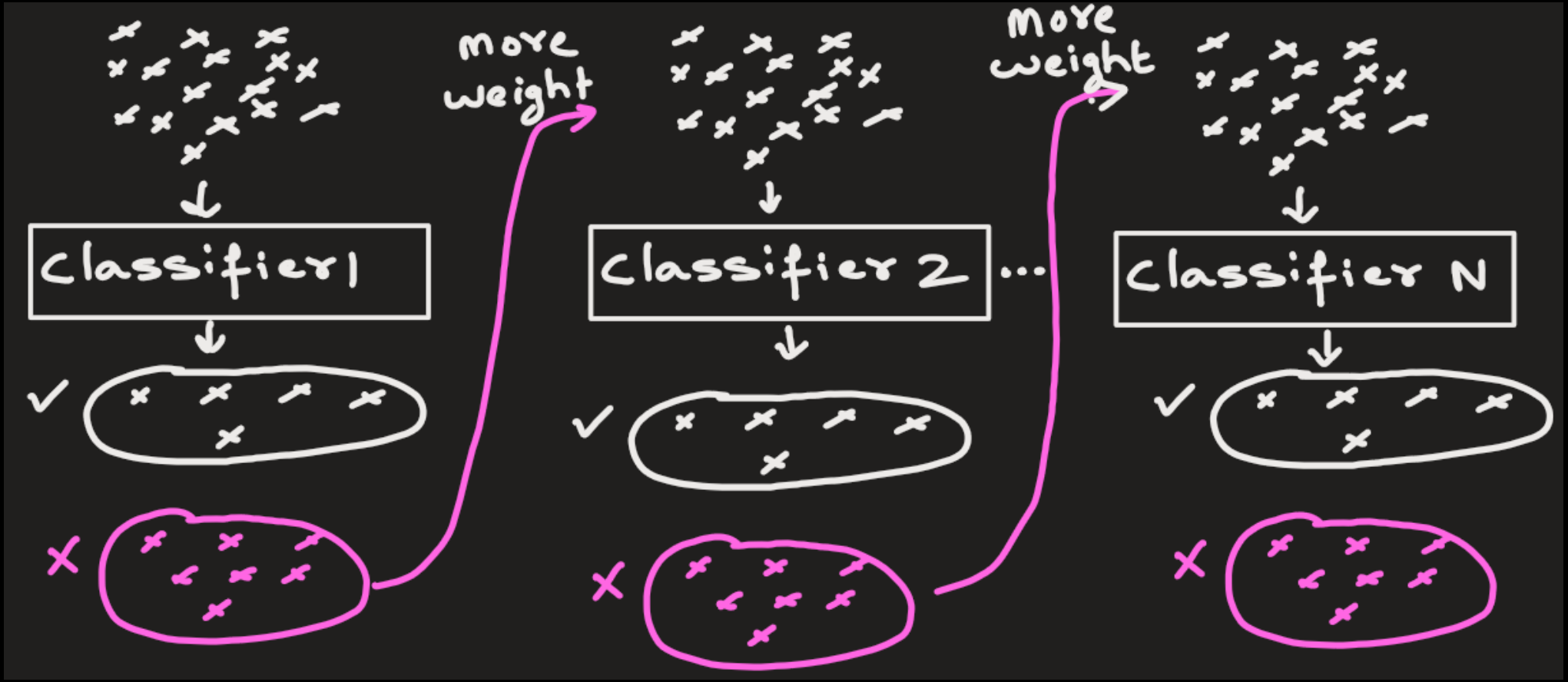
Key Characteristics

- Models are trained **one after another**
- Later models depend on earlier ones



Popular Boosting Algorithms:

- AdaBoost
- Gradient Boosting
- XGBoost / LightGBM / CatBoost



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